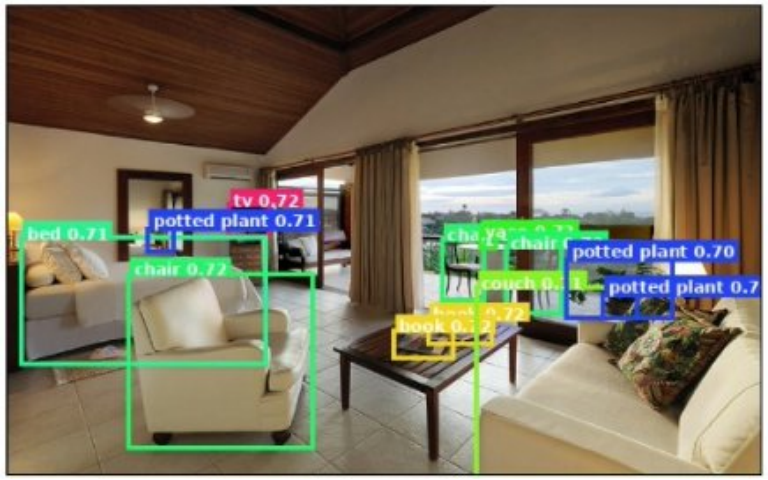
A plug-and-play shield against adversarial attacks

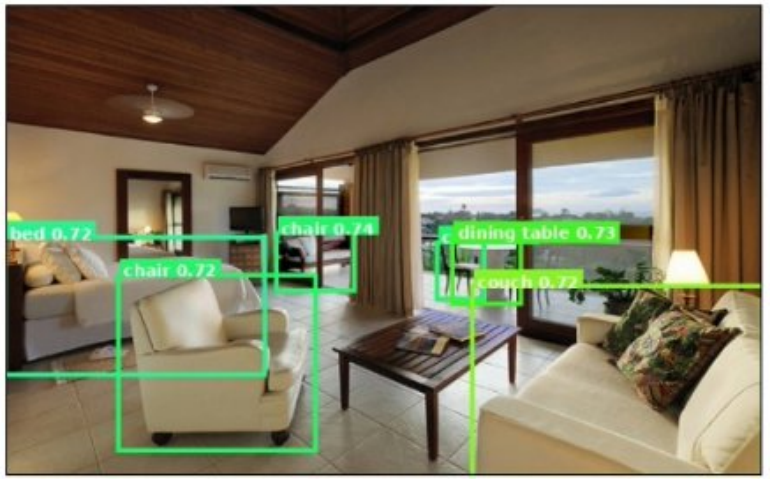
Training-free input filters in front of any frozen vision model, tested on Florence-2 and YOLO-World over all 5,000 COCO val images. Where we are, and the options we are weighing.

Research discussion · October 2026

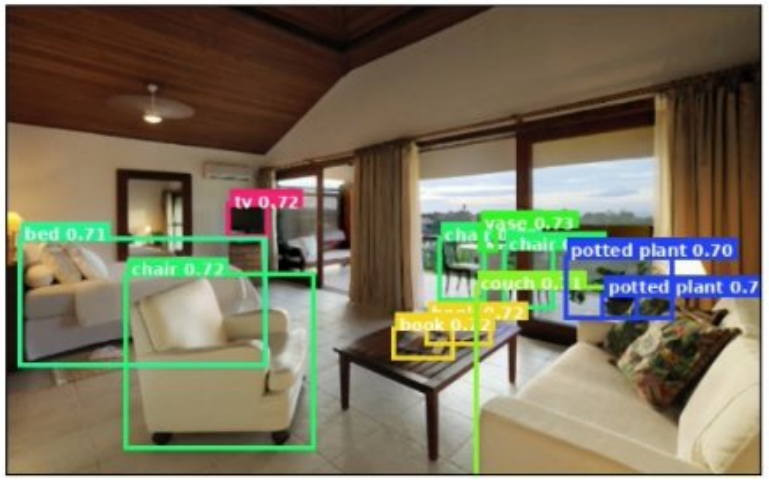
Can a plug-in filter block undo adversarial attacks?



Clean · 12 detections



PGD attack (ε = 0.03) · 6 detections



Our 3-filter block · 11 detections

The attack

Invisible noise (at most 8/255 per pixel), computed from the model's own gradients.

Our block

Filters in front of the model. No fine-tuning, no extra network: attach it to any model.

Our study

Which filters, alone or combined, protect which model, task and attack?

How we got here: four phases



Florence-2 under FGSM and PGD, first filter defenses

Published: IEEE APSCON 2026

Screened 8 filters and 4 chained pipelines (Florence, 1,000 images)

Kept TVM, JPEG, Median; chains with NLM or resizing failed

Added YOLO-World and OCR: 2 models × 3 attacks = 9 settings

5 filters + 3 parallel ensembles, 1,000 images

Full survey: 20 defenses on all 5,000 COCO val images

Submitted to an ECCV 2026 workshop

Clean-image cost and error bars measured; adaptive attack built

GPU runs next

- **Lesson that shaped the block:** chaining filters in series lost accuracy, so we run them in parallel and merge after the model.
- **Set aside early:** DiffPure and SmoothVLM (built for classification and text). DiffPure comes back as option B.

Setup: 2 models, 3 attacks, 20 training-free defenses

Models (frozen)

- Florence-2-base: VLM for detection and OCR
- YOLOv8x-worldv2: open-vocabulary detector

Data

- COCO val2017, all 5,000 images
- YOLO-World prompted with the 80 COCO classes; Florence labels mapped to COCO

Attacks (white-box)

- FGSM: 1 step, ε = 0.03
- PGD: 10 steps, random start
- Patch: 35×35 at the centre, 100 Adam steps

Defenses

- 13 filters: JPEG, Median, Gaussian, TVM, Bilateral, NLM, BM3D, …
- 7 ensembles of 3–4 filters, merged after the model

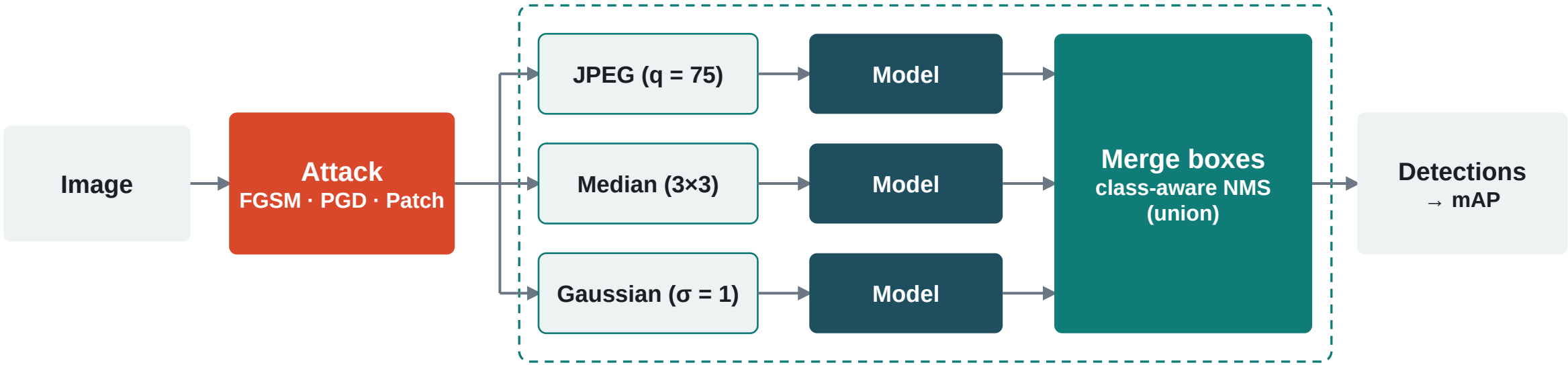
Metrics

- Detection: COCO mAP@[.5:.95]
- OCR: similarity to the model's own clean output

Attacker

- So far: blind, knows the model but not the block
- Next: adaptive, knows the block too (section 3)

How the block works: filter, run, merge



Our block (EnsJMG shown): the model itself is untouched

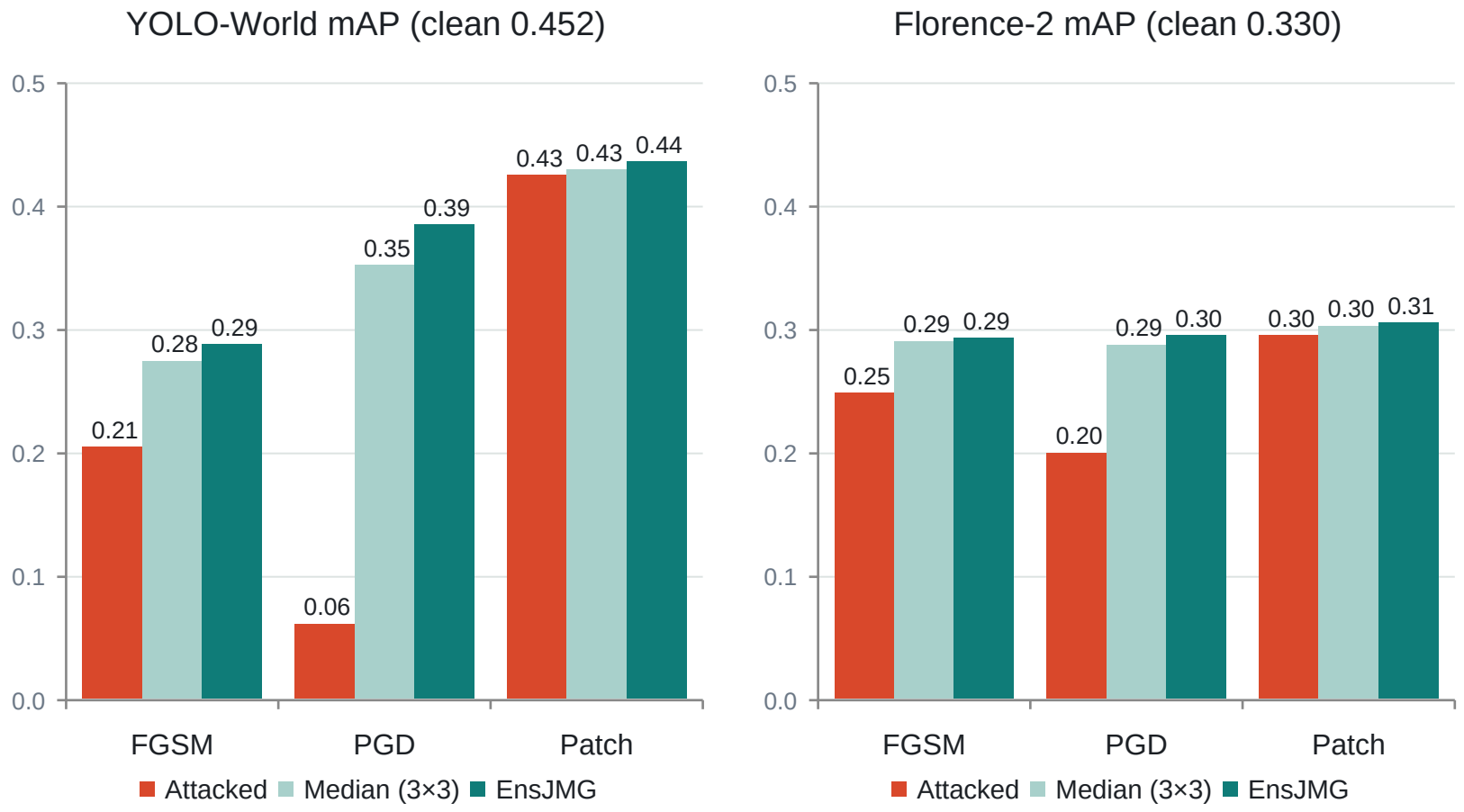
Plug-and-play

No training and no change to the model: the same block runs in front of Florence-2 and YOLO-World.

Union merge

The model runs once per branch (3×). A box survives if any branch keeps it, so one clean view is enough.

Blind attacks: the block recovers most of the loss



N = 5,000 images, mAP@[.5:.95]. OCR, not shown: PGD drops similarity to 0.29; the best filters restore 0.65–0.66.

Takeaways

1. PGD hits hardest: YOLO falls from 0.45 to 0.06 mAP
2. Under PGD, EnsJMG restores 85% of clean mAP on YOLO and 90% on Florence
3. Median and all 7 ensembles helped in all 9 settings

Everything we tried: what helped where

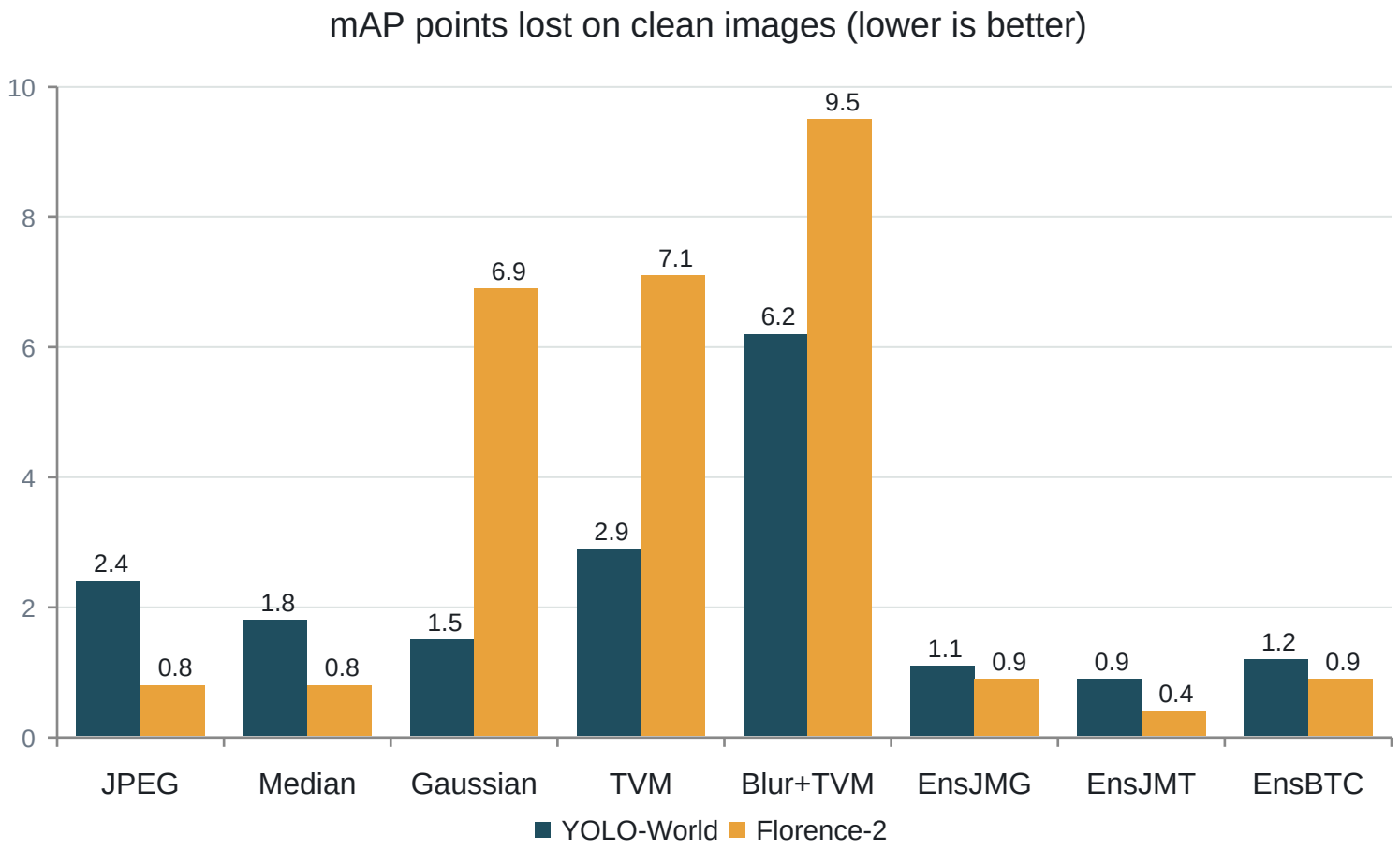
Defense	YOLO-World · mAP			Florence-2 · mAP			Florence-2 · OCR			Helped
	FGSM	PGD	Patch	FGSM	PGD	Patch	FGSM	PGD	Patch	
Ens4W	**+8.5**	**+33.5**	+1.0	+4.1	**+9.2**	+0.6	**+21.5**	**+35.8**	**+18.3**	**9/9**
EnsBTC	**+10.8**	**+33.1**	+0.6	+4.3	**+9.5**	+0.6	**+15.6**	**+29.2**	**+16.7**	**9/9**
EnsJMT	**+7.8**	**+32.8**	+0.9	+4.4	**+9.6**	+0.9	**+20.9**	**+35.5**	**+5.9**	**9/9**
EnsBMT	**+6.9**	**+32.4**	+0.9	+4.4	**+9.3**	+1.4	**+18.4**	**+35.7**	+4.4	**9/9**
EnsJMG	**+8.3**	**+32.3**	+1.1	+4.5	**+9.5**	+1.0	**+17.6**	**+32.6**	**+19.5**	**9/9**
EnsJBT	**+6.1**	**+32.2**	+0.4	+4.5	**+9.5**	+1.5	**+19.1**	**+36.7**	+1.8	**9/9**
EnsJTS	**+5.7**	**+31.3**	+0.8	+3.0	**+7.4**	+0.4	**+16.1**	**+28.6**	+2.2	**9/9**
Median (3×3)	**+7.0**	**+29.1**	+0.4	+4.2	**+8.8**	+0.7	**+10.6**	**+25.2**	**+19.8**	**9/9**
Gaussian (σ = 1)	**+9.0**	**+32.5**	+0.0	+1.3	**+6.2**	−2.4	**+18.9**	**+31.5**	**+18.9**	8/9
JPEG (q = 75)	+4.5	**+27.6**	−1.2	+4.3	**+9.3**	+0.9	**+17.6**	**+32.2**	+2.4	8/9
Bilateral	+2.2	**+24.1**	−0.5	+4.5	**+9.5**	+2.1	**+13.6**	**+34.3**	−0.3	7/9
Anisotropic diff.	+1.0	**+23.9**	−1.7	+3.7	**+8.5**	+0.7	**+16.1**	**+33.6**	−1.1	7/9
TVM	**+7.1**	**+32.3**	−2.4	−0.5	+4.0	−4.6	**+25.2**	**+36.0**	+2.3	6/9
Blur+TVM	**+12.1**	**+32.0**	−4.2	−0.2	+4.5	−5.2	**+12.4**	**+19.3**	**+9.4**	6/9
Dithering	+1.4	**+9.6**	−1.2	+1.2	**+5.2**	−1.1	**+9.1**	**+22.2**	−4.0	6/9
SVD (low-rank)	+0.0	+0.2	+0.0	−0.3	−0.2	−0.4	+0.4	+1.5	+0.2	6/9
Bit-depth (4-bit)	+1.4	**+7.6**	−1.1	−1.7	+1.9	−4.8	**+9.9**	**+22.7**	−3.4	5/9
NLM	−0.3	**+11.7**	−0.3	−0.7	+0.5	−1.7	+2.1	**+8.9**	−1.6	4/9
BM3D	–	–	–	–	–	–	**+22.3**	**+30.4**	−2.3	2/3
Random Resize	–	–	–	–	–	–	**+18.6**	**+29.0**	**+18.4**	**3/3**

≥ +5 points 0 to +5 0 to −5 ≤ −5

Change vs. attacked, in points (×100); N = 5,000, one run each

Shaded rows: ensembles; letters name the filters (J JPEG, M Median, G Gaussian, T TVM, B Bilateral, S SVD): Ens4W = J+M+T+G, EnsBTC = J + Blur+TVM + M. BM3D and Random Resize: OCR only (Random Resize rescales the image, so its boxes are not comparable).

Plug-and-play needs a low price on clean images



N = 1,000 images (our earlier runs), 95% paired bootstrap CIs. Florence: the CIs of JPEG, Median and all three ensembles include zero.

≈ 1 point

what an ensemble costs on clean YOLO images; on Florence, too small to measure

6.9 → 0.9

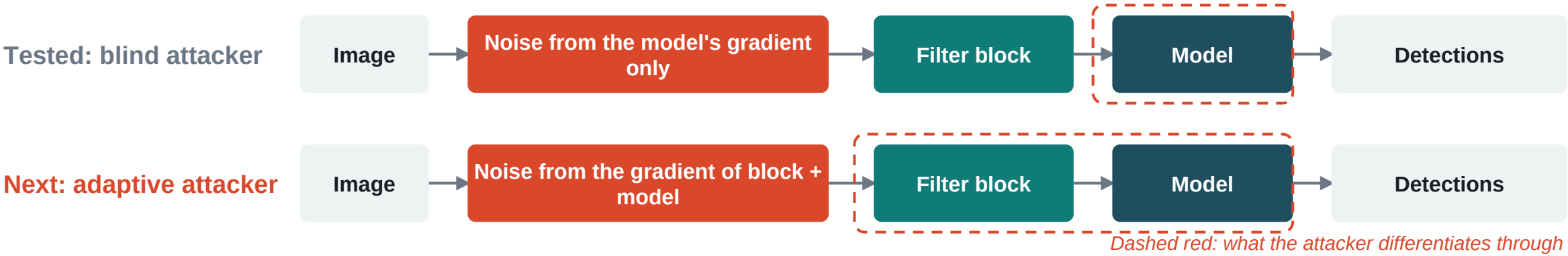
Gaussian alone costs Florence 6.9 points; inside EnsJMG the merge brings that down to 0.9

What the workshop reviewers asked, and what we did

EmergingAD workshop at ECCV 2026: not accepted, scores 4/10 and 6/10 (reviewer confidence 3 and 2). Main request: test an attacker who knows the defense.

Reviewers asked	What we did	Status
Test an attacker who knows the defense	Adaptive attack (BPDA + EOT) aimed at the merge; seeded, 3 seeds	**Code ready · GPU next**
Report the cost on clean images	Measured for every defense, with 95% confidence intervals	**Done**
Add error bars	Paired bootstrap CIs on every comparison; seeded attacks for new runs	**Done**
State clear research questions	Paper rebuilt around four questions: cost, dependence on model and task, combining, adaptivity	**Drafted**
OCR score is consistency, not accuracy	Ground-truth text from COCO-Text (option D)	**Open**
Fit to driving and physical threats	Choose a venue that fits a general robustness study	**To decide**

Next test: an attacker who knows the block

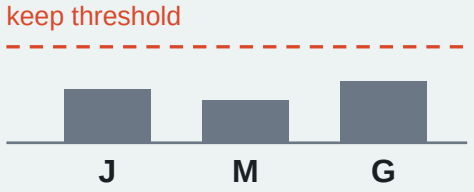


Fool the branches on average



Gaussian branch still sees it: box kept

Fool the strongest branch (ours)



All branches below: box removed

How the adaptive attack works

- **BPDA:** pass gradients through JPEG and Median, which have none
- **EOT:** optimise one noise over all branches of the block
- **Our twist:** one clean branch keeps the box, so push down the strongest branch, not the average
- **Closest prior work:** AdROD (arXiv 2608.16031, Aug 2026) uses the average

Illustration: one object's confidence in each branch after the attack

Can the block handle it? Not tested yet

Why it may hold

- The attacker must beat all three branches at once
- JPEG and Median have no gradient; the attacker must approximate them
- Strong blind results: 85–90% of clean mAP under PGD

Why it may not

- BPDA + EOT broke JPEG-, TVM- and bit-depth-style defenses on classifiers (Athalye et al., ICML 2018)
- Ensembles of weak defenses were not strong (He et al., WOOT 2017)

Possible outcomes, all publishable

Holds

A training-free shield that survives an adaptive attacker on detectors: a strong positive result

Partly holds

Report how much blind testing overstates it, then harden the block

Breaks

An evaluation finding: blind tests overstate input filters for detectors, by a measured amount

If it falls, training-free ways to harden it

- **Randomize** filter settings per image (raises the cost; EOT can still average over it)
- **Add a branch:** a pretrained diffusion purifier (option B)
- **Raise an alarm** when branches disagree, as in feature squeezing (Xu et al., NDSS 2018)

Option B: add a modern pretrained denoiser

DiffPure

Nie et al., ICML 2022

Add noise, then denoise with a pretrained diffusion model. The standard purifier today; as a baseline or as a 4th branch. Slow per image, and hard to attack fairly (Lee & Kim, ICCV 2023).

Plug-and-play: yes

HGD

Liao et al., CVPR 2018

A denoiser trained against adversarial noise, guided by the model's own features. Strong, but needs training for each model.

Plug-and-play: no

Generic denoisers

DnCNN, Restormer, NAFNet

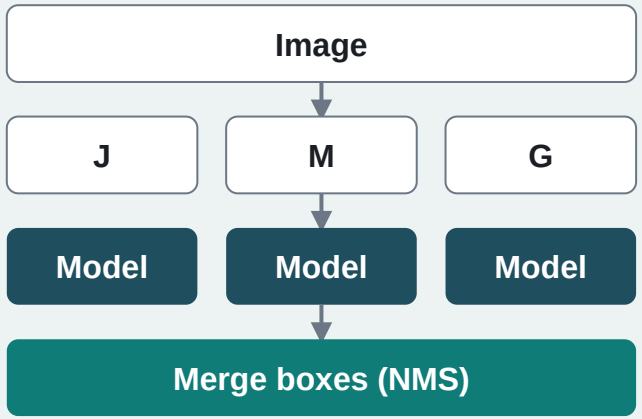
Pretrained, but on random noise, not adversarial noise. The HGD paper shows leftover noise gets amplified inside the network.

Plug-and-play: yes

Already tested (classic, all training-free): JPEG, Median, Gaussian, TVM, Blur+TVM, Bilateral, anisotropic diffusion, NLM, BM3D, bit-depth, dithering, SVD, random resize.

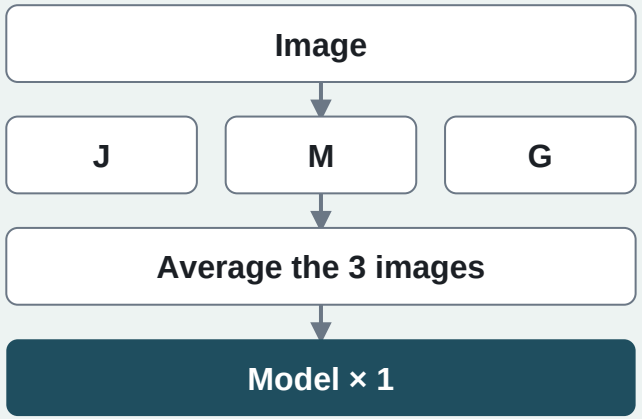
Option C: one filter instead of three?

Ensemble (now)



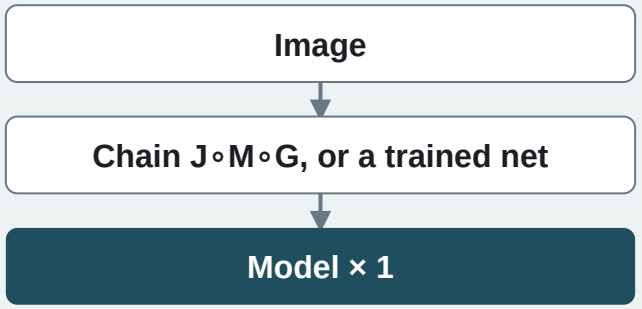
3 model passes; the merge happens after the model

Pixel average



1 model pass; a cheap ablation: does merging boxes matter?

Single filter



Chaining already tried (Blur+TVM): it cost Florence 9.5 points on clean images

Our view: no exact single equation (the merge is after the model; Median and JPEG are non-linear). Planned: the pixel average at one pass, and a control that merges three plain views (flip, rescale) to separate the merge's effect from the filters'. A trained single filter would be smooth, so easier to attack.

Options side by side

Option	What it adds	Compute	Main risk	Our ranking
A · Adaptive attack on the merge	Answers the main review; our novelty (strongest-branch attack)	~5–7 h, one A6000 (estimate)	Close preprints (arXiv 2610.29196)	**Must do**
B · Diffusion purifier	Modern, still training-free; baseline or 4th branch	High: slow per image	Fair adaptive test is hard	**Strong add**
C · Pixel average	The block at one model pass; is it the merge or the filters?	Low	Small, supporting claim	**Ablation**
D · OCR ground truth	Accuracy, not only consistency (COCO-Text labels)	Low	Coverage of val2017 unknown	**Nice to have**
E · Hardening	Randomized filters, branch-disagreement alarm	Low	Needed only if A breaks it	**If needed**

Venue

- TMLR: rolling, judges correctness over novelty
- CVPR / ICCV 2027 workshops: need one sharp finding
- Main conference only if A is striking

Constraints

- Must clearly extend our APSCON 2026 paper
- GPU runs need the full COCO val2017 and an A6000
- TMLR no longer takes surveys: frame it as an evaluation

Where we would value your view

1. **Novelty**
   Is a fair adaptive test of the block enough of a contribution, or do reviewers expect a new defense?

2. **If it breaks**
   Which training-free fix is most promising: randomized filters, a diffusion branch, or an alarm?

3. **Baselines and data**
   Which modern baseline is a must (DiffPure, HGD, other), and is COCO enough?

4. **Venue**
   TMLR, a CVPR / ICCV 2027 workshop, or a main conference?

Backup

Full detection and OCR tables · ensemble vs Median · clean cost with CIs · attack details

BACKUP

Full detection results, N = 5,000

Defense	YOLO-World (mAP)			Florence-2 (mAP)		
	FGSM	PGD	Patch	FGSM	PGD	Patch
Clean	**0.4519**	**0.4519**	**0.4519**	**0.3300**	**0.3300**	**0.3300**
Attacked	**0.2051**	**0.0619**	**0.4257**	**0.2488**	**0.2005**	**0.2958**
Median (3×3)	**0.2750**	**0.3525**	**0.4299**	**0.2910**	**0.2881**	**0.3032**
Gaussian (σ = 1)	0.2948	0.3871	0.4261	0.2621	0.2625	0.2719
JPEG (q = 75)	0.2503	0.3377	0.4133	0.2922	0.2937	0.3044
TVM	0.2757	0.3847	0.4017	0.2440	0.2410	0.2494
Blur+TVM	0.3265	0.3815	0.3840	0.2467	0.2457	0.2437
Bilateral	0.2268	0.3024	0.4206	0.2938	0.2956	0.3165
Anisotropic diff.	0.2151	0.3006	0.4085	0.2855	0.2856	0.3028
Dithering	0.2191	0.1581	0.4139	0.2610	0.2529	0.2852
SVD (low-rank)	0.2055	0.0638	0.4259	0.2462	0.1990	0.2920
Bit-depth (4-bit)	0.2189	0.1382	0.4150	0.2321	0.2191	0.2477
NLM	0.2018	0.1793	0.4225	0.2420	0.2058	0.2791
Ens4W	0.2896	0.3969	0.4358	0.2897	0.2927	0.3016
EnsBTC	0.3134	0.3931	0.4317	0.2915	0.2955	0.3014
EnsJMT	0.2832	0.3902	0.4347	0.2931	0.2961	0.3051
EnsBMT	0.2739	0.3857	0.4347	0.2931	0.2938	0.3099
EnsJMG	**0.2882**	**0.3853**	**0.4366**	**0.2934**	**0.2957**	**0.3057**
EnsJBT	0.2664	0.3834	0.4298	0.2942	0.2953	0.3105
EnsJTS	0.2617	0.3751	0.4333	0.2792	0.2745	0.2994

Highlighted: Median, EnsJMG; ensembles shaded. Random Resize omitted (it rescales the image, so its boxes are not comparable); BM3D was run on OCR only.

BACKUP

Full OCR results (Florence-2), N = 5,000

Defense	FGSM	PGD	Patch	Defense	FGSM	PGD	Patch
Attacked	**0.346**	**0.294**	**0.383**	EnsJBT	0.537	0.661	0.401
TVM	0.598	0.654	0.406	Ens4W	0.561	0.652	0.566
Bilateral	0.482	0.637	0.380	EnsBMT	0.530	0.651	0.427
Anisotropic diff.	0.507	0.630	0.372	EnsJMT	0.555	0.649	0.442
JPEG (q = 75)	0.522	0.616	0.407	EnsJMG	0.522	0.620	0.578
Gaussian (σ = 1)	0.535	0.609	0.572	EnsBTC	0.502	0.586	0.550
BM3D	0.569	0.598	0.360	EnsJTS	0.507	0.580	0.405
Random Resize	0.532	0.584	0.567				
Median (3×3)	0.452	0.546	0.581				
Bit-depth (4-bit)	0.445	0.521	0.349				
Dithering	0.437	0.516	0.343				
Blur+TVM	0.470	0.487	0.477				
NLM	0.367	0.383	0.367				
SVD (low-rank)	0.350	0.309	0.385				

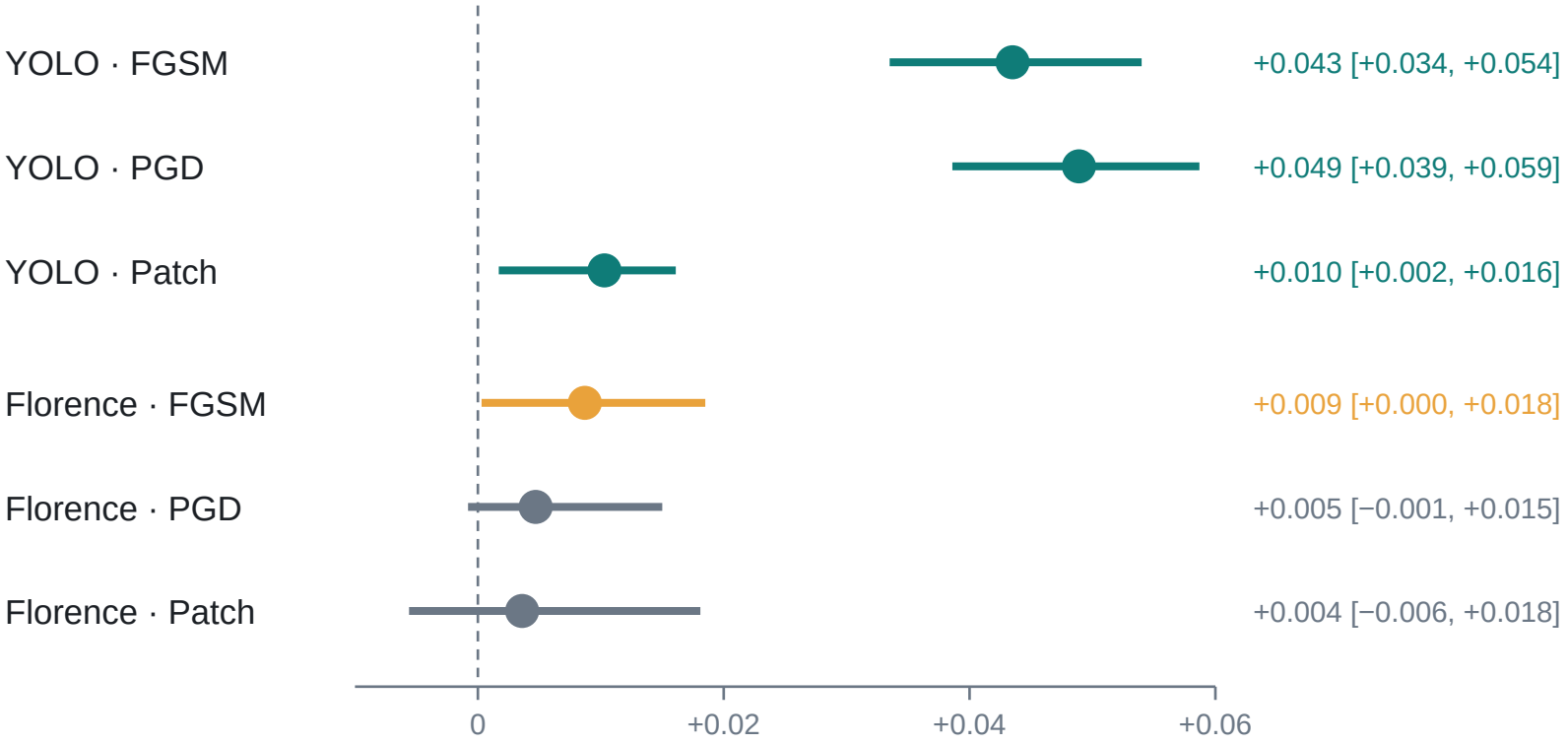
Similarity of the OCR text to the model's own clean output (character-level; clean = 1.0 by construction). Sorted by PGD. Left: filters; right: ensembles.

BACKUP

Is the ensemble worth 3× compute over Median?

Best ensemble minus Median, mAP with 95% CI

N = 1,000 images · paired bootstrap ● clearly above 0 ● borderline ● within noise



- **YOLO:** yes, +0.04 to +0.05 mAP under FGSM and PGD
- **Florence:** the gain is within noise, so Median alone (one pass) may be enough
- **For option C:** a cheaper block may suit some models

BACKUP

Clean-image cost with 95% CIs

Defense	YOLO-World: Δ mAP [95% CI]	Florence-2: Δ mAP [95% CI]
JPEG	−0.024 [−0.033, −0.015]	−0.008 [−0.021, +0.004]
Median	−0.018 [−0.025, −0.010]	−0.008 [−0.021, +0.007]
Gaussian	−0.015 [−0.024, −0.007]	−0.069 [−0.085, −0.056]
TVM	−0.029 [−0.038, −0.021]	−0.071 [−0.088, −0.060]
Blur+TVM	−0.062 [−0.073, −0.050]	−0.095 [−0.115, −0.081]
EnsJMG	−0.011 [−0.018, −0.002]	−0.009 [−0.020, +0.008]
EnsJMT	−0.009 [−0.015, −0.002]	−0.004 [−0.015, +0.011]
EnsBTC	−0.012 [−0.020, −0.004]	−0.009 [−0.020, +0.009]

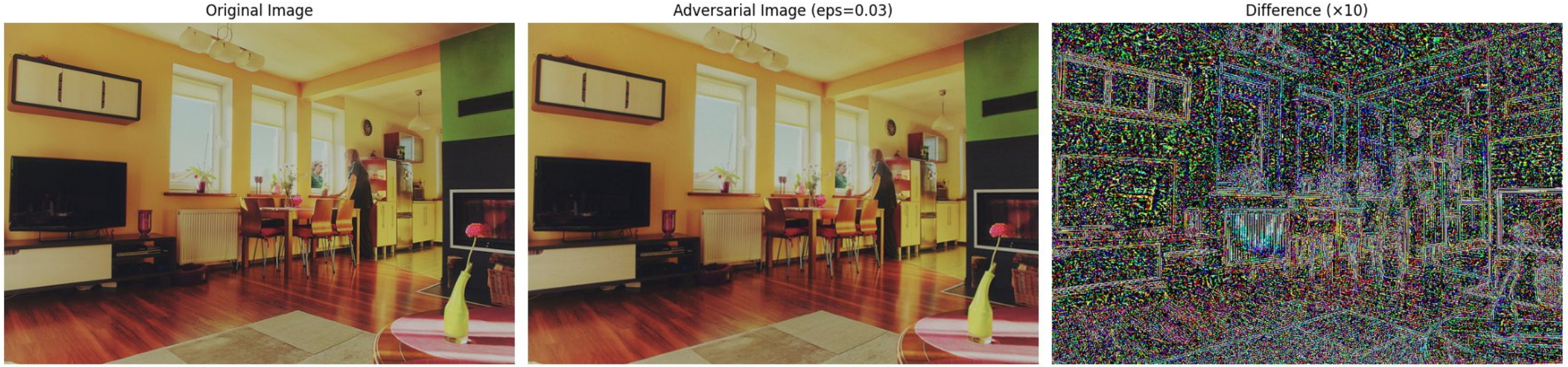
CI below zero: a measurable cost

CI includes zero: no measurable cost

Defended minus undefended mAP on clean images, N = 1,000, paired bootstrap (B = 1,000).

BACKUP

Attack details and budgets



FGSM example from our paper: original, adversarial (ε = 0.03), difference ×10.

Florence-2

- Loss: token cross-entropy against its own clean output
- Normalized space: ε = 0.03 ≈ 1.7/255 in pixels
- PGD step α = ε/4 (5,000-image runs)

YOLO-World

- Loss: lower the top-class confidence at all 8,400 anchors
- Raw [0, 1]: ε = 0.03 = 7.65/255, ~4.4× Florence's budget
- Patch: 35×35 at the centre, 100 Adam steps, lr 0.02